Semiconductor Solar Cell Devices – Physics, Fabrication, and Limits

GROUP-7

Inception of solar cell physics

The photovoltaic effect causes solar cells to convert the light into electrical energy that can be used to power modern gadgets. The first step is absorption: when the energy of the incident photon is greater than or equal to the bandgap of the semiconductor ($h\nu \geqslant E_g$), an electron of the valence band gets excited to the conduction band. This energizes the atom's electrons.

When illuminated continuously, the system does not remain thermally equilibrated anymore. The Fermi level splits into quasi-Fermi levels. This gradient acts as the cause of the electrical current to arise. The internal electric field available within the depletion area sweeps these carriers at the p-n junction to separate these generated carriers. The carriers are transported by the minority carriers that diffuse to the edge of the depletion region and drift across the depletion region due to the electric field. The cell's electrical behavior in the fourth quadrant is given by the equation $I=IL-I0[exp(qV/kT)-1]$. The overall efficiency is measured with Short Circuit Current (Jsc), Open Circuit Voltage (Voc) and Fill Factor (FF).

Manufacturing and Fabrication Processes

The process of making a crystalline silicon solar cell involves a number of precisely-controlled physical and chemical steps. These are generally grouped into three main phases.

The cutting through the process has left saw damage on the original silicon wafer (generally p type <100>) helps to remove this damage. To get rid of this damage, about 10 to 20 micrometers of material is etched off and the surface is prepared for a better light reflection. This is followed by alkaline texturing, which creates random pyramidal structures on the surface through an anisotropic etching process with a height of 2 to 3.5 micrometers. This

texturing is very efficient and lowers the Weighted Average Reflection from about 30% to 10-12%.
The creation of a p-n junction takes place in stage II by creating a n-type emitter on p-type bulk silicon. This is achieved via tube diffusion at a temperature range of 800 to 900°C using $POCl_3$ as source. The dopant diffuses into silicon as a result of the reaction. The process produces phosphosilicate glass, or PSG, from which is removed with hydrofluoric acid, or HF.

Once the junction is completed, silicon nitride which is hydrogenated is deposited by means of a PECVD. This layer has a dual purpose. This type of coating is one that utilizes destructive interference to trap more light. The film hydrogen passes allows free electrons to passivate dangling silicon bonds on the surface, reducing the Surface Recombination Velocity drastically.

In stage III, electrical contacts are applied via screen printing on to the silicon wafer. A silver (Ag) paste is utilized for printing the front to form busbars and grids whereas a full-area aluminum (Al) paste is applied to the back. RTP of wafer is done at more than 1000 °C. As a result of this firing, an Al-Si eutectic alloy is formed at the back which then results in a highly doped p-type BSF. As the internal electric field that this BSF produces repels the minority carrier electrons away from the rear surface, it minimizes recombination losses and hence causes a notable increase in Open Circuit Voltage.

Loss Analysis and Efficiency Constraints

Even optimized fabrication of solar cells cannot eliminate certain optical and resistive losses.

The total short circuit current losses due to optical factors are about 7.38 mA/cm^2. The rear aluminum contact parasite absorption dominates and causes a loss of approximately 3.62 mA/cm^2 in this case. The front metal grid's physical presence also creates shading that blocks about 1.2 mA/cm^2. Front ARC reflection of incident light and escape reflection from emission of light also cause small optical loss. When we consider the electrical effects, the main reason behind Fill Factor loss is series resistance contributing to approximately 1.53% loss while shunt resistance is negligible for fully optimized cells.

Recombination is the Main Killer of Efficiency in Solar Cells. Carriers diffuse when they are generated before being collected. These mechanisms reduce the diffusion length: $L = D\tau$; if smaller than cell thickness, they are lost as heat before extraction. There are three different types.

The Shockley-Read-Hall (SRH) concept differs from both radiative recombination and Auger recombination. Like where it is manifested. If due to impurities and grain boundaries it is predominantly limits to the multi-crystalline silicon cells.

Radiative Recombination is the basic reverse process of absorption of light. It’s dictated by thermodynamics and is completely unavoidable.

Auger recombination is a three-particle process which becomes the dominant loss mechanism in the heavily doped regions, like the emitter and the Back Surface Field.

Future Architectures and Theoretical Limits.

The maximum efficiency of a single-junction solar cell is defined by the Shockley-Queisser Limit, a thermodynamic constraint. This limitation is set by two essential unavoidable losses. The first is spectral mismatch, in which lower energy photons ($h\nu<Eg$) pass through the cell completely not being absorbed and thus yield 0% power. The photons with energy greater than the bandgap are absorbed upon thermalization but the excess energy is lost to the lattice as heat. The most dominant mechanism of loss in photovoltaic physics is thermalization. Due to these constant thermodynamic limits, single-junction crystalline silicon (that has a bandgap of 1.1 eV) has a fundamentally limited maximum theoretical efficiency at about 33%.

Technologies for Thin Film Multi-Junctions To get rid off the 33% barrier, engineers stack various semiconductor materials in multi-junction architectures, making the

Solar cell/large scale – Power plant solar cell different components of the solar spectrum, which they can absorb. An upper layer of InGaP (energy bandgap = 1.9 eV) absorbs UV and blue light. A middle layer of InGaAs (1.4 eV) absorbs visible light. Finally, a lower layer of Germanium (0.67 eV) absorbs infrared. The gap reduces thermalization likelihood. The layers are connected via low-resistance tunnel junctions. Thanks to this architecture, lab efficiencies over 46% under concentrated light have been achieved. However, this architecture is complex, requiring exact lattice matching and current matching (the lowest current limits the whole string).

Currently, crystalline silicon dominates the market with a **90%** share, but second-generation thin films account for the remaining **10%**. Thin-film materials like Cadmium Telluride (CdTe) and CIGS feature direct bandgaps with massive absorption coefficients (>105cm−1). This physical advantage means they require only about **1 micrometer** of material to absorb sunlight, compared to the **150 micrometers** needed for indirect bandgap silicon.

Conclusion

As the industry moves forward, continuous engineering is required to push devices closer to their fundamental physical limits. Current efficiency stands at roughly **25%** for silicon and **18-22%** for thin films, while multi-junctions exceed **40%**. Future directions rely heavily on advanced passivation architectures like PERC and PERT to further mitigate rear surface recombination. Furthermore, placing emerging materials like Perovskites as top cells in tandem with silicon bottom cells presents the most viable path to finally breaching the 33% Shockley-Queisser limit in commercial applications. Ultimately, while physics strictly dictates the thermodynamic limits, ongoing engineering and material science continue to close the gap to perfection